

EXPERIMENT: USE CASE DIAGRAM DESIGN

1. Aim

To design and understand a UML Use Case Diagram representing the interaction between different actors and the **NEXO** system for event updates, registration, and QR-based entry verification.

2. Objective

- To illustrate the functional requirements of the system
- To identify the actors interacting with the system
- To understand how students and admins interact with the platform
- To represent event registration and QR-based entry verification
- To visualize the relationship between event updates, registration, and notifications
- To study system behavior for event participation and management

3. Definition of Use Case Diagram

A Use Case Diagram is a type of UML (Unified Modeling Language) diagram that represents the functional behavior of a system from the user's perspective.

It shows:

- Different users (actors) interacting with the system
- Functionalities provided by the system
- Relationships between actors and use cases
- Include and extend relationships between system functions
- Overall system interaction in a simple visual form

Use Case Diagrams are widely used to represent system requirements and user interactions.

4. Procedure

Step 1: Identify Actors

List all participating actors:

- Student
- Admin
- Event Scanner (Staff/Volunteer)

Step 2: Define the System

The main system considered:

NEXO – Campus Event Management and QR-Based Registration System

The system allows students to view events, register, receive QR codes, and enables admins to manage events and verify participation.

Step 3: Draw System Boundary

- Draw one large rectangle
- Label it as **NEXO System**
- This represents the complete system boundary

Step 4: Add Use Cases

Inside the system boundary, add oval-shaped use cases:

- Login / Registration
- View Events
- Register for Event
- Generate QR Code
- View Registered Events
- Receive Notifications
- Create Event
- Update/Delete Event
- Manage Registrations
- Scan QR Code
- Verify Entry
- Dashboard Monitoring

Step 5: Connect Actors to Use Cases

Student connects to:

- Login / Registration
- View Events
- Register for Event
- Generate QR Code
- View Registered Events
- Receive Notifications

Admin connects to:

- Login / Registration
- Create Event
- Update/Delete Event
- Manage Registrations
- Dashboard Monitoring

Event Scanner connects to:

- Scan QR Code
- Verify Entry

Step 6: Add Include Relationships

Use <<include>> relationships for:

- Register for Event → Generate QR Code
- Scan QR Code → Verify Entry
- Create Event → Send Notifications

Step 7: Finalize Diagram

- Use proper stick figures for actors
- Use oval shapes for use cases
- Use straight connectors
- Maintain clear spacing and neat alignment
- Ensure proper UML structure and readability

5. Notations Used in the Diagram

Notation	Meaning
Stick Figure	Represents actors like User, Admin
Rectangle Box	Represents system boundary
Oval Shape	Represents system functionality
Solid Line	Association between actor and use case
<<include>>	Mandatory included functionality
Labels	Describe use case names

6. Procedure to Create Use Case Diagram in Draw.io

Step 1: Open Draw.io

Step 2: Create New Diagram → Blank Diagram

Step 3: Enable UML → UML Use Case

Step 4: Add Actors:

- Student
- Admin
- Event Scanner

Step 5: Draw system boundary:

- Label it as **NEXO System**

Step 6: Add use cases inside the boundary

Step 7: Connect actors to use cases

- Add <<include>> relationships using dashed arrows

Step 8: Final formatting:

- Align properly
- Maintain spacing
- Keep diagram clean

7. Procedure to Upload Diagram in Azure DevOps Wiki

Step 1: Open Azure DevOps → Go to your project

Step 2: Navigate to Wiki section

Step 3: Open/Create page (e.g., *Use Case Diagram*)

Step 4: Click **Edit**

Step 5: Upload image:

- Drag & drop OR Insert Image

Step 6: Adjust size and alignment

Step 7: Click **Save / Save & Close**

(Add commit message like “*Added Use Case Diagram*”)

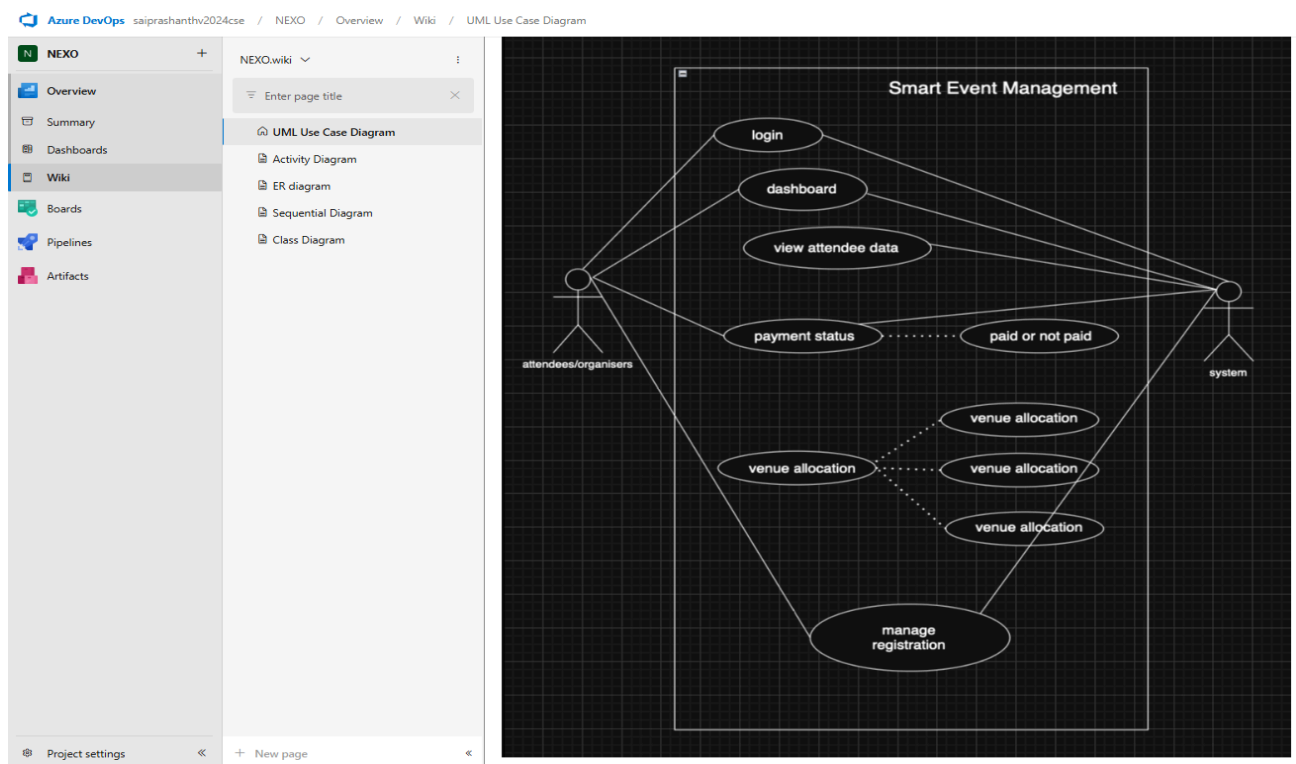
Step 8: Verify:

- Ensure diagram is visible
- Check clarity and readability

OUTCOME:

The Use Case Diagram was successfully created to represent the interaction between students, admins, and event scanners in the **NEXO** system.

It clearly illustrates event browsing, registration, QR code generation, notification delivery, and entry verification. The diagram effectively demonstrates how the system ensures smooth event management and participation through a structured and user-friendly workflow.



8. Result

The Use Case Diagram was successfully designed for the **NEXO** project. It clearly represents system functionalities, actor interactions, and event management processes such as event browsing, registration, and QR-based entry verification.

The diagram helps in understanding the complete workflow of the system, from event updates to participation, and supports better system analysis and effective project presentation.